FNO$^{\mathbf{i}\theta}$: Extended Fourier Neural Operator for learning state and optimal control of distributed parameters systems

***Abstract*—**

I. INTRODUCTION

II. PROBLEM FORMULATION

Let P denote a matrix with entries $P_{j,k}, 1 \leq j \leq p, 1 \leq l \leq q$. Each entry $P_{j,l}$ is a polynomial function in $\mathbb{C}^n$. Let $D_j = -\mathbf{i}\partial/\partial_{\xi_j}, j = 1, \ldots, n$, and $D = (D_1, \ldots, D_n)$. Then, $P(D)$ is a matrix of linear partial differential operators with constant coefficients. Following this notation, we consider the system of inhomogeneous PDEs in a convex open subset Ω of $\mathbb{R}^n$ of the form,

$$P(D)\phi = u, \tag{1}$$

or, more explicitly,

$$\sum_{l=1}^{q} P_{j,l}(-\mathbf{i}\partial/\partial_{\xi_1}, \ldots, -\mathbf{i}\partial/\partial_{\xi_n})\phi_l = u_j, \quad j = 1, \ldots, p, \tag{2}$$

where ϕ is the state and u is the control input. In this paper, we focus on smooth functions u_j and ϕ_l in $\mathcal{L}^2(\Omega) \cap \mathcal{C}^\infty(\Omega), j = 1, \ldots, p$ and $l = 1, \ldots, q$. Note that the system (1) has solutions in more general spaces, such as the space of distributions [1].

Remark 1: In practice, the domain Ω can either denote a spatial domain or a product of a spatial domain and a time interval, i.e., $\Omega = \Omega_x \times \Omega_t$ where $\Omega_x \subseteq \mathbb{R}^{n-1}$ and $\Omega_t \subseteq \mathbb{R}_+$. For brevity, we use a single variable ξ to denote a point in Ω and do not distinguish between spatial and temporal components.

Remark 2: Appropriate boundary conditions are typically specified to ensure the uniqueness of the solution to the system (1). We focus on general solutions to (1) first and postpone the discussion of boundary conditions to numerical experiments.

Example 1: Throughout the paper, we will use the following one-dimensional heat equation as a running example to illustrate the main results.

$$\frac{\partial \phi(x,t)}{\partial t} = \frac{\partial^2 \phi(x,t)}{\partial x^2} + u(x,t). \tag{3}$$

In this example, P is a single polynomial and $P(\xi) = \mathbf{i}\xi_1 + \xi_2^2$.

The optimal control problem under consideration is to find a control input u that minimizes a cost functional $J(\phi, u)$ of the following form subject to the system (1),

$$J(\phi, u) = \int_\Omega |\phi(\xi) - \phi_d(\xi)|^2 + \lambda |u(\xi)|^2 d\xi, \tag{4}$$

where ϕ_d is a given desired state and $\lambda > 0$ is a regularization parameter. When $\phi_d = 0$, this problem is a special case of the classical linear quadratic regulator (LQR) problem for PDEs [2]. For $\Omega = \Omega_x \times \Omega_t$, where $\Omega_x = \mathbb{R}^{n-1}$ and $\Omega_t = \mathbb{R}_+$, explicit expressions for optimal control in the unbounded spatial domain have been derived in [3]. The one-dimensional case where Ω_x is a finite interval has also been studied in [4]. However, explicit expressions for optimal control in a general (potentially bounded) domain Ω are not readily available. This is relevant for many practical applications where the spatial domain is typically bounded.

In this paper, we propose to develop a neural operator for learning explicit expressions for optimal control. The key to our approach is to leverage an integral representation of optimal control in designing neural operator architecture. The integral representation is derived in the following section.

III. INTEGRAL REPRESENTATIONS FOR OPTIMAL CONTROL

To derive an integral representation for optimal control, we first present the first-order optimality conditions that are satisfied by any optimal control to (4) subject to (1). Then, we leverage the first-order optimality conditions and the Ehrenpreis fundamental principle [1], [5] to show that any optimal control can be represented as an integral in $\mathbb{C}^n$.

For the LQR problem under consideration, the first-order optimality conditions can be expressed as a system of coupled PDEs using the adjoint approach. The following result follows by specializing [6, Section 1.6.3] to the LQR problem.

Lemma 1: Let u^* be an optimal control for (4) subject to (1) and ϕ^* be the corresponding optimal state. Then, there exists an adjoint state ψ^* such that the following system of equations is satisfied,

$$\begin{cases} P(D)\phi^* = u^*, \\ P(D)^\dagger \psi^* = -(\phi^* - \phi_d), \\ u^* = \frac{1}{\lambda}\psi^*, \end{cases} \tag{5}$$

where $P(D)^\dagger$ denotes the adjoint of $P(D)$.

Substituting the last equation for u^* into the first equation in (5), we observe that optimal control u^* and state ϕ^* can be represented as a solution to the following system of PDEs,

$$\begin{bmatrix} P(D) & -I \\ P(D)^\dagger & \lambda I \end{bmatrix} \begin{bmatrix} \phi^* \\ u^* \end{bmatrix} = \begin{bmatrix} 0 \\ \phi_d \end{bmatrix}. \tag{6}$$

Note that (6) is a system of linear PDEs with constant coefficients and can be written in a similar form as (1).

In the following, we first present the existence of a solution to (1) in a convolution form. Then, we combine this convolution solution with the Ehrenpreis-Palamodov fundamental principle [5], [7] to derive an integral representation for optimal control.

Lemma 2 (Theorem 2.11 in [8]): There exists a distribution $\tilde{E}$ such that $P(D)\tilde{E} = \delta$, where δ is the Dirac delta function. Moreover, the convolution of $\tilde{E}$ with the control input u, i.e., $\tilde{\phi} = \tilde{E} * u$, is a solution to (1).

Remark 3: The classical FNO is inspired by the existence of a solution $\tilde{\phi}$ in a convolution form. The key ingredient in FNO is to learn the convolution operator in the Fourier space, which reduces to learning a multiplication parameter.

The existence of $\tilde{\phi}$ does not necessarily imply that every solution to (1) can be represented in a convolution form. On the other hand, the Ehrenpreis-Palamodov fundamental principle states that any solution to (1) with $u = 0$ can be represented as an integral of exponential functions multiplied by polynomials over the complex domain.

Lemma 3 ([5], [7], [9]): Let $\bar{\phi}$ denote a solution to the system (1) with $u = 0$. Then, there exists zero sets of polynomials $\bar{\mathcal{V}}_j \subset \mathbb{C}^n, j = 1, \dots, s$, polynomials $\bar{Q}_{j,l}, j = 1, \dots, s, l = 1, \dots, m_j$, and measures $d\bar{\mu}_{j,l}$ whose support lies in $\bar{\mathcal{V}}_j$, such that $\bar{\phi}$ can be represented as an integral of the following form,

$$\bar{\phi} = \sum_{j=1}^{s} \sum_{l=1}^{m_j} \int_{\bar{\mathcal{V}}_j} e^{\mathbf{i}k\cdot\xi} \bar{Q}_{j,l}(\xi, k) d\bar{\mu}_{j,l}(k). \tag{7}$$

Moreover, the function $e^{\mathbf{i}k\cdot\xi}\bar{Q}_{j,l}(\xi, k)$ satisfies the system of PDEs in (1) for each $k \in \bar{\mathcal{V}}_j$.

Applying Lemmas 2–3 to the coupled state-control system (6), we obtain the following result.

Theorem 1: Let u^* denote an optimal control to (4) subject to (1) and ϕ^* denote the corresponding optimal state. Then, there exists a distribution E, zero sets of polynomials $\mathcal{V}_j \subset \mathbb{C}^n, j = 1, \dots, s$, functions $Q_{j,l}, j = 1, \dots, s, l = 1, \dots, m_j$, and measures $d\mu_{j,l}$ whose support lies in $\mathcal{V}_j$, such that u^* and ϕ^* can be represented in the following form,

$$\begin{bmatrix} \phi^* \\ u^* \end{bmatrix} = E * \begin{bmatrix} 0 \\ \phi_d \end{bmatrix} + \sum_{j=1}^{s} \sum_{l=1}^{m_j} \int_{\mathcal{V}_j} e^{\mathbf{i}k\cdot\xi} Q_{j,l}(\xi, k) d\mu_{j,l}(k). \tag{8}$$

Moreover, the function $e^{\mathbf{i}k\cdot\xi}Q_{j,l}(\xi, k)$ satisfies the system of PDEs in (6) for each $k \in \mathcal{V}_j$.

Proof: According to Lemma 2, there exists a distribution E such that

$$\begin{bmatrix} P(D) & -I \\ P(D)^\dagger & \lambda I \end{bmatrix} E * \begin{bmatrix} 0 \\ \phi_d \end{bmatrix} = \begin{bmatrix} 0 \\ \phi_d \end{bmatrix}. \tag{9}$$

Combining (6) and (9), we have

$$\begin{bmatrix} P(D) & -I \\ P(D)^\dagger & \lambda I \end{bmatrix} \left(\begin{bmatrix} \phi^* \\ u^* \end{bmatrix} - E * \begin{bmatrix} 0 \\ \phi_d \end{bmatrix} \right) = 0. \tag{10}$$

Applying Lemma 3 to (10), we obtain the integral representation in (8). ∎

Example 2: The integral representation in (8) is quite abstract and non-constructive. We illustrate an example of this representation for the one-dimensional heat equation (3) given the initial condition $\phi(x, 0) = \phi_0(x)$. Let $\phi_d = 0, \lambda = 1$. Following [3, Section V-B], the optimal state and control for the heat equation can be represented as follows,

$$\begin{aligned} \phi^*(x,t) &= \int_{\mathbb{R}} e^{\mathbf{i}k_x x - \omega(k_x)t} \hat{\phi}_0(k_x) \frac{dk_x}{2\pi}, \\ u^*(x,t) &= \int_{\mathbb{R}} e^{\mathbf{i}k_x x - \omega(k_x)t} (k_x^2 - \omega(k_x)) \hat{\phi}_0(k_x) \frac{dk_x}{2\pi}, \end{aligned} \tag{11}$$

where $\omega(k_x) = \sqrt{k_x^4 + 1}$ and $\hat{\phi}_0(k_x)$ is the Fourier transform of the initial condition $\phi_0(x)$. Let $k = \begin{bmatrix} k_x & k_t \end{bmatrix}$. Then, the expression (11) can be represented in the form of (8) by letting $s = 1, m_1 = 1, \mathcal{V}_1 = \{k \in \mathbb{C}^2 : -k_t^2 + k_x^4 + 1 = 0\}$, $Q_{1,1}(x, t, k) = \begin{bmatrix} 1 & k_x^2 + k_t \end{bmatrix}^\top$, $d\mu_{1,1}(k) = \frac{1}{2\pi}\hat{\phi}_0(k_x)dk$ if $k_t \in \mathbb{R}_+, k_x \in \mathbb{R}$ and $d\mu_{1,1}(k) = 0$ otherwise, where dk is the Lebesgue measure in $\mathbb{R}^2$.

For the heat equation over $\mathbb{R}$, it is sufficient to represent optimal control as an integral over $\mathbb{R}$ as in (11). Note that the integrals in (11) can also be deformed to integrals over contours in $\mathbb{C}$. For the heat equation over a finite interval, real domain is not enough and optimal control needs to be represented as an integral over well-constructed contours in $\mathbb{C}$ in the form (8), see [4].

IV. Extended Fourier Neural Operator

In this section, we first present the architecture of the classical Fourier Neural Operator (FNO). Then, we propose an extended FNO architecture, called FNO$^{\mathbf{i}\theta}$, that is inspired by the complex integral representation in (8). When the phase θ is zero, FNO$^{\mathbf{i}\theta}$ reduces to the classical FNO.

A. Classical FNO

Let a, b denote an input and output function in Banach spaces $\mathcal{A}(\Omega; \mathbb{R}^{n_a}), \mathcal{B}(\Omega; \mathbb{R}^{n_b})$, respectively. In this paper, we restrict our attention to the case where $\mathcal{A}$ and $\mathcal{B}$ are subspaces of $\mathcal{L}^2(\Omega) \cap \mathcal{C}^\infty(\Omega)$, respectively. Following [10], [11], we consider a neural operator $\mathcal{N}(a) : \mathcal{A} \to \mathcal{B}$ of the form

$$\mathcal{N}(a) = \mathcal{Q} \circ \mathcal{L}_L \circ \mathcal{L}_{L-1} \circ \cdots \circ \mathcal{L}_1 \circ \mathcal{R}(a), \tag{12}$$

where L is a given length in $\mathbb{N}$, $\mathcal{R} : \mathcal{A}(\Omega; \mathbb{R}^{n_a}) \to \mathcal{B}(\Omega; \mathbb{R}^{n_v})$ is a lifting operator with $n_v \geq n_b$, $\mathcal{Q} : \mathcal{B}(\Omega; \mathbb{R}^{n_v}) \to \mathcal{B}(\Omega; \mathbb{R}^{n_b})$ is a projection operator, and $\mathcal{L}_l : \mathcal{B}(\Omega; \mathbb{R}^{n_v}) \to \mathcal{B}(\Omega; \mathbb{R}^{n_v}), l = 1, \dots, L$ are nonlinear operator layers. The lifting and projection operators are typically linear transformations implemented using multilayer perceptrons [12], and the nonlinear operator layer is of the form,

$$\mathcal{L}_l(v_l)(\xi) = \sigma\left(W_l v_l(\xi) + (\mathcal{K}_l v_l)(\xi)\right), \tag{13}$$

where $v_l \in \mathcal{B}(\Omega; \mathbb{R}^{n_v})$ is the output from the previous layer, σ is a nonlinear activation function applied component-wise, $W_l \in \mathbb{R}^{n_v \times n_v}$ is a weight matrix, and $\mathcal{K}_l : \mathcal{B}(\Omega; \mathbb{R}^{n_v}) \to \mathcal{B}(\Omega; \mathbb{R}^{n_v})$ is a bounded linear operator.

Different types of neural operators vary by the choice of the linear operator $\mathcal{K}_l$. To learn solution operators of PDEs, the linear operator $\mathcal{K}_l$ in FNO is defined as a convolution

operator in the Fourier space motivated by the existence of a solution in a convolution form by Lemma 2. Since the convolution operation is transformed to a multiplication in the Fourier space, the operator $\mathcal{K}_l$ in FNO is expressed as follows,

$$(\mathcal{K}_l v_l)(\xi) = \int_{\mathbb{R}^n} e^{\mathbf{i}k\cdot\xi} \hat{K}_l(k)\hat{v}_l(k)\frac{dk}{2\pi}, \qquad (14)$$

where $\hat{K}_l(k) \in \mathbb{C}^{n_v \times n_v}$ is a matrix indexed by frequency variable k, $\hat{v}_l(k)$ is the Fourier transform of v_l.

Remark 4: In practice, the value of input function a is typically only available at a finite number of discretized spatial locations in Ω. Also, the matrix $\hat{K}_l(k)$ is usually learned for a finite number of frequencies k in the Fourier space. Therefore, FNO is implemented in a discretized manner, where (14) is approximated as a finite sum. We refer readers to [10]–[12] for more details on the discretized version of FNO and other practical implementation issues.

In summary, the parameters to be learned in FNO include the weight matrices W_l and the Fourier space parameters $\hat{K}_l(k)$ for all layers $l = 1, \ldots, L$. Despite the motivation from linear PDEs, FNO has been successfully applied to learning solution operators of all different types of nonlinear PDEs, with applications in various fields [10]–[12]. This is partially due to the expressiveness of a neural operator architecture in (12), especially when the number of layers L (length) and the lifted dimension n_v (width) are sufficiently large. In the following, we introduce an extended FNO architecture, called FNO$^{\mathbf{i}\theta}$, which will be shown to be more expressive than the classical FNO when n_v is small in numerical experiments.

B. Extended FNO$^{\mathbf{i}\theta}$

The key to the design of FNO is the convolution operator $\mathcal{K}_l$ in (14) represented in the Fourier space, inspired by the existence of a solution in a convolution form, see Lemma 2. However, not every solution to (1) can be represented in a convolution form, which could potentially limit the expressiveness of FNO. In Lemma 3 and Theorem 1, we show that any state and optimal control of (1) can be represented in a form involving a complex integral, see (7)–(8). Comparing (8) to (14), the integrands all contain the leading exponential term $e^{\mathbf{i}k\cdot\xi}$. This motivates us to design an extended FNO architecture, where the linear operator $\mathcal{K}_l$ in (13) is given by

$$(\mathcal{K}_l^\theta v_l)(\xi) = \int_{\mathbb{R}^n} e^{\mathbf{i}z(k,\theta_l)\cdot\xi} \hat{K}_l(k)\hat{v}_l(k)\frac{dk}{2\pi}, \qquad (15)$$

where $\theta_l = (\theta_{l,1}, \ldots, \theta_{l,n}), z_j(k,\theta_l) = k_j e^{\mathbf{i}\theta_{l,j}}, j = 1, \ldots, n$. In other words, the original frequency variable k is the modulus of the new frequency variable z, and we introduce a new parameter θ_l to be learned as the phase of the complex variable z. Clearly, when the phase θ_l is zero, $\mathcal{K}_l^\theta$ in (15) reduces to $\mathcal{K}_l$ in (14).

In the following section, we present numerical experiments to show that introducing the new parameter θ greatly improves the expressiveness of the neural operator, especially for a small value of the lifted dimension n_v for which the classical FNO may fail.

V. Numerical Experiments

VI. Conclusions and Future Works

A. Conclusions

B. Future Works

VII. Acknowledgments

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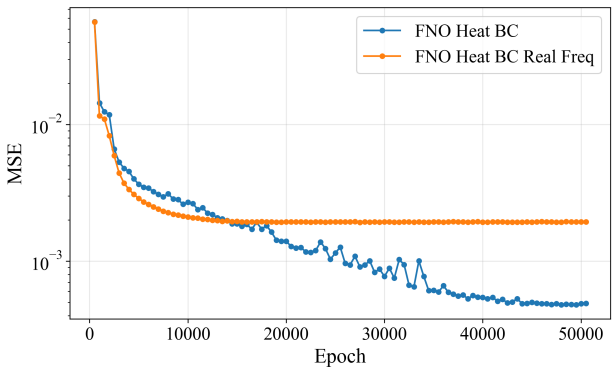


Fig. 1: Comparison of the original MSE history between FNO Heat BC and FNO Heat BC Real Freq.

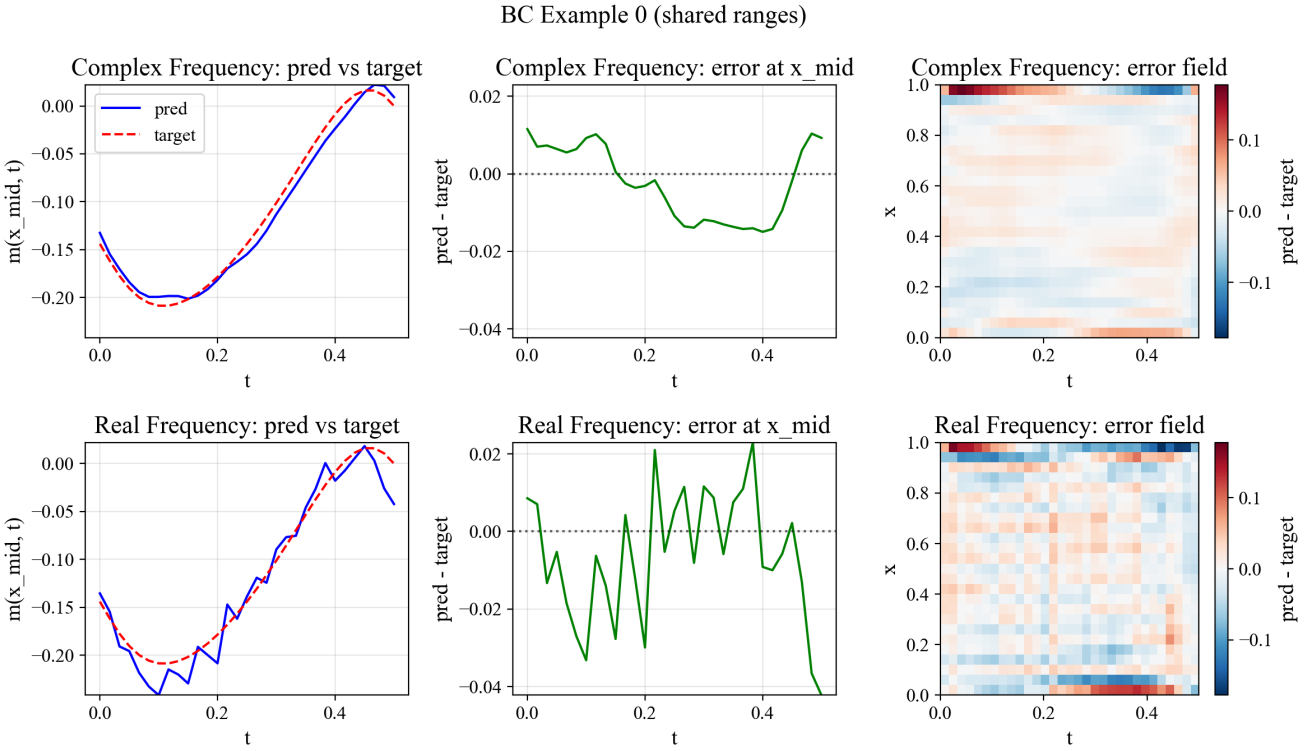


Fig. 2: Numerical results for the one-dimensional heat equation.

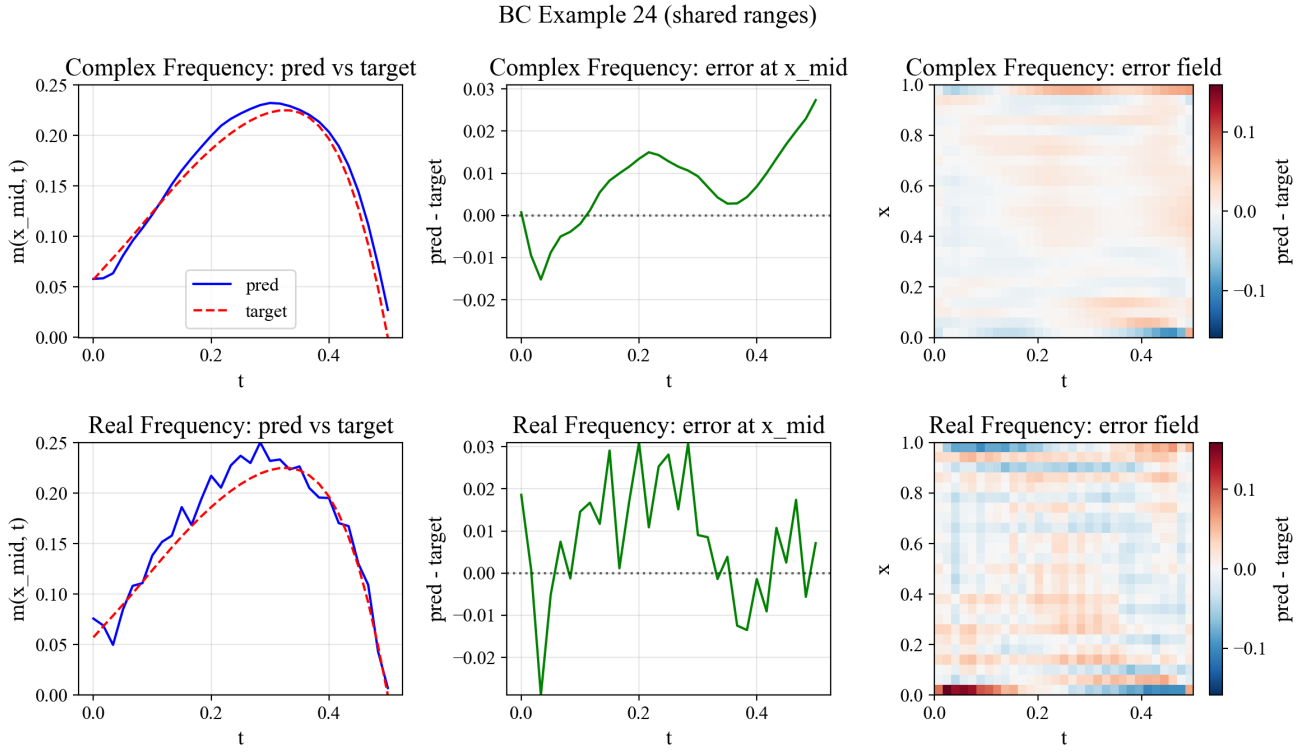


Fig. 3: Numerical results for the one-dimensional heat equation.

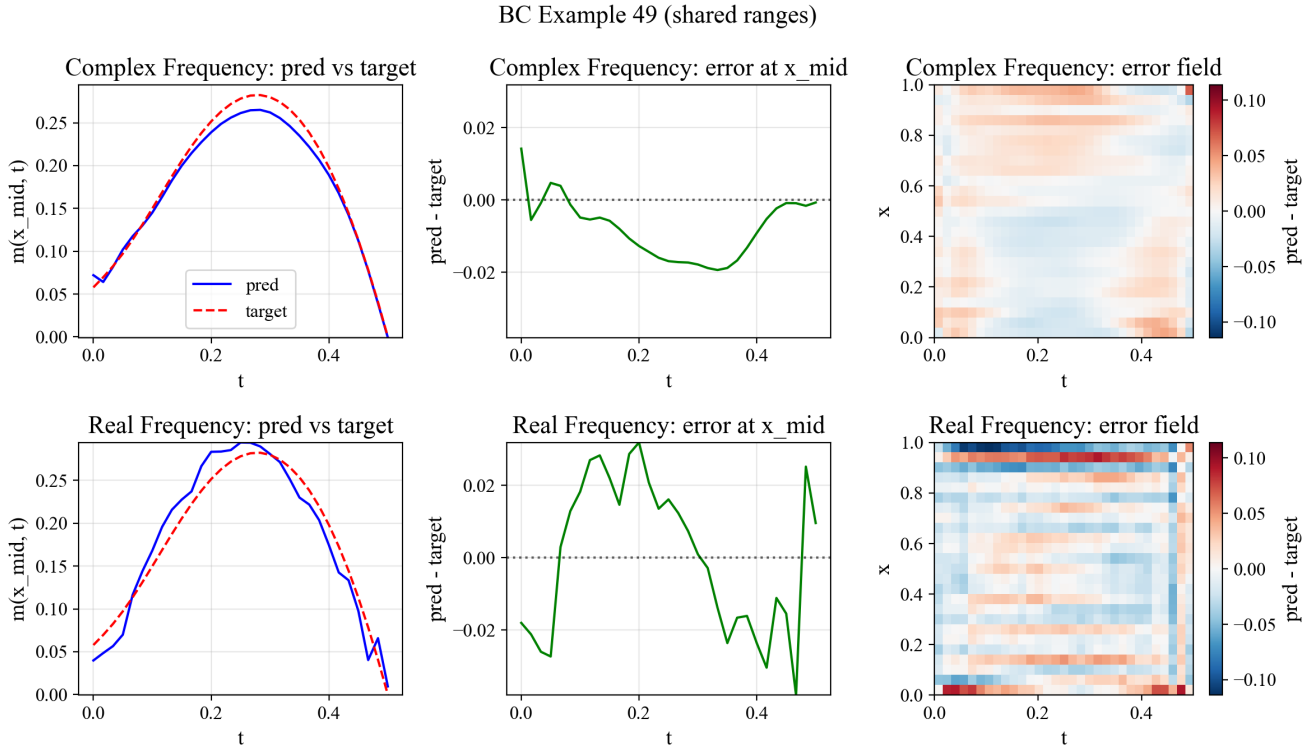


Fig. 4: Numerical results for the one-dimensional heat equation.